

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Full-stack engineer with production ownership of **TypeScript, Node.js, React, and PostgreSQL** -- shipped 10+ user-facing features at an AI startup from schema design to deployed UI. Built **Elasticsearch-powered search pipelines** and consumer-facing real-time platforms. 6+ hackathon wins; builder who ships fast and takes ownership end-to-end.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Owned full backend of production platform -- shipped 10+ customer-facing features using **TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Redis, Auth0, Docker, and Sentry**; real-time event pipelines and REST APIs serving production users across deployments.
- Built multi-model AI agent tooling across 3 providers and 8+ models (Claude, GPT-4o, Gemini) via MCP; document processing pipelines (PDF gen, email parsing, S3); reviewed PRs across the engineering team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation (OCR, NLP, ML) improving accuracy 25%; automated AWS and GCP deployments with Grafana + Sentry observability.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Consumer-facing platform built with **NestJS + Next.js + React**: MySQL outbox pattern, MongoDB TTL feed projections, **Redis sorted sets for sub-ms reads**, Server-Sent Events for live score updates. OpenTelemetry + Prometheus observability; benchmarked with k6; GitHub Actions CI/CD.
TypeScript, NestJS, Next.js, React, MySQL, MongoDB, Redis, Docker, Prometheus

VectorLift -- Semantic Search & Ranking Engine 2026

Production semantic search on MS MARCO: **BM25 baseline (Elasticsearch)**, dense bi-encoder retrieval (sentence-transformers), cross-encoder re-ranker. Kafka-driven indexing pipeline for event-based corpus ingestion; **6 pipeline configs** evaluated with NDCG@10, MRR@10, and bootstrap significance testing via FastAPI + Streamlit dashboard.
Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Streamlit, Docker

Inventory Reservation Service 2026

Transactional inventory API in Java 21 + Spring Boot preventing overselling via **PostgreSQL row-level locking (SELECT FOR UPDATE)**. Idempotent POST endpoints, RFC 9457 errors, Flyway migrations, Spring Security. Concurrency verified with Testcontainers: two simultaneous reservations against qty 1 produce exactly one success + one 409 Conflict.
Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker, GitHub Actions

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Algorithms, Data Science, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, DS & Algorithms, Big Data Infra, Data Warehousing

SKILLS

- Languages:** TypeScript, Python, JavaScript, Java, SQL, C
- Stack:** React, Next.js, NestJS, Node.js, Express, Prisma ORM, Tailwind, REST APIs, Temporal.io, Kafka
- Data:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant, Docker, GitHub Actions, CI/CD
- AI/ML:** OpenAI API, Claude API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, MCP, LLM Evals
- Awards:** HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)